

Responsible AI Statement

AI-Powered Malaria Detection — Ngao Labs Capstone Proof of Concept

1. Who does this system affect?

This proof of concept affects three groups. First, users of the model's output include laboratory technologists, clinicians and other healthcare workers who could use an AI prediction as decision-support when examining blood-cell images. Second, the people the model makes predictions about are patients whose blood-cell images are being classified, even though they may never know that an AI system was involved. Third, Kenyan patients and frontline laboratory communities—particularly those in rural, peri-urban and high-malaria-burden settings—are affected but have no formal voice in this POC's design. The training dataset consists of 27,558 labelled microscopic cell images from the publicly available NIH/NLM Malaria Cell Images dataset, rather than Kenyan clinical images.

2. What is the worst plausible outcome?

The most serious plausible outcome is that a patient with a genuinely parasitized blood cell is incorrectly classified as uninfected, leading a healthcare worker to delay or withhold appropriate malaria treatment. This is not hypothetical model risk: in the current test set, 310 parasitized cells were classified as uninfected, producing a parasitized-class recall of 77.50%. The system must therefore remain decision-support and must never be presented as a standalone clinical diagnosis.

3. Who bears the cost of errors?

The cost of an incorrect prediction is primarily borne by the patient, not by the developer or organisation operating the model. A false negative can mean delayed treatment; a false positive can result in unnecessary clinical attention or treatment. This asymmetry means monitoring cannot focus only on overall accuracy. The model should be monitored especially for recall of parasitized cases, false-negative rates and subgroup performance, with human clinical judgement remaining the final decision authority.

4. What redress mechanism exists?

The POC currently has no dedicated clinical appeal mechanism because it is not deployed as a clinical decision system. If developed for real-world use, users should be able to request human review, report an incorrect prediction, obtain an explanation of how the result was generated, and have incorrect personal information corrected. Kenya's Data Protection Act provides data-subject rights and addresses solely automated decisions; where applicable, a data subject can request reconsideration or a decision that is not based solely on automated processing. Kenya's Data Protection (General) Regulations also require safeguards around automated decision-making, including prevention of errors, mitigation of discriminatory effects and access to human intervention.

5. How will the model be monitored?

For a future deployment, I would conduct monthly performance monitoring, with additional review after any major change in data, imaging equipment or clinical workflow. The primary metric would be recall for the parasitized class, supported by precision, F1-score, false-negative rate and subgroup performance. I propose 80% parasitized recall as the minimum deployment target: if recall falls below 80%, the model should be investigated and retrained; if it falls below 70%, the model should be withdrawn from operational use until validated. This threshold is a proposed safety gate for the POC, not a clinically validated standard.

Fairness criterion: Equal Opportunity

The fairness criterion selected is Equal Opportunity because the most consequential error in this screening context is failing to identify a genuinely parasitized case. Equal Opportunity focuses on achieving comparable true-positive rates across relevant groups. However, this POC cannot yet demonstrate demographic Equal Opportunity because the available dataset does not provide the Kenyan demographic, geographic or clinical metadata required for meaningful subgroup comparison. The criterion should therefore become a formal validation requirement before Kenyan deployment.

Disaggregated findings

The available class-level evaluation is shown below. The Uninfected class falls well below the proposed 80% recall threshold. This is a significant finding: although the model detects many parasitized cells, it incorrectly labels a large number of uninfected cells as parasitized. The confusion matrix records 939 such false positives.

Datasheet summary

The dataset represents 27,558 pre-segmented, labelled microscopic blood-cell images classified as parasitized or uninfected, originating from the NIH/NLM dataset. It does not adequately represent the diversity of Kenyan patients, healthcare facilities, microscopy equipment, geographic settings or full-slide clinical workflows; the project therefore identifies Kenyan clinical validation as a necessary next step.

Disaggregated Evaluation Table

Subgroup	N	Recall	F1-Score
Parasitized cells	13,779	0.7750	0.6310
Uninfected cells	13,779	0.3186	0.4128

Overall assessment: This POC demonstrates technical feasibility, not clinical readiness. Responsible deployment requires stronger validation, representative Kenyan data, human oversight, transparent redress, and continuous monitoring before the system is used to influence patient care.